

Coating Analyzer Binek 2.4 Beta – User Manual

Professional Customer Support Guide

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1. Introduction

Coating Analyzer Binek 2.4 Beta is an advanced tool for material image analysis, designed for research laboratories, quality control, and teaching. It allows:

- Calculating porosity (pore area fraction) on the whole image or within selected regions (ROI).
- Advanced binarization methods: Otsu, Multi-Otsu, adaptive, manual thresholding, fusion voting, watershed, and more.
- Linear (length) and area measurements.
- Hardness measurement using Vickers or Knoop methods.
- Scale calibration (μm , mm, inches).
- Exporting results to Word (.docx) and Excel (.xlsx) reports with charts and images.
- Live USB camera preview and frame capture.

The application runs as a standalone `.exe` file – **no Python installation or additional libraries are required.**

2. First Launch and Basics

After launch, the main window appears with:

- **Left panel** – image view tabs (Live, Clipboard, Original, Processed, Thickness, ROI, Binary, Fusion).
- **Right panel** – control tabs (Camera, Adjustments, Thickness, Porosity, Hardness, Calibration).
- **Bottom status bar** – cursor coordinates, calibration status, zoom slider, and "Fit to Window" button.

Recommended workflow:

1. Load an image (file, clipboard, or camera).
2. Perform calibration (if scale is known).
3. (Optional) Create ROI(s).
4. Adjust brightness, contrast, gamma, and filters in the **Adjustments** tab.
5. Choose a binarization method (Porosity tab) and adjust its parameters.
6. Click **Whole Image** (or **Selected ROI**) to compute porosity.
7. Perform length or hardness measurements.

8. Save results (Word/Excel report) and images.
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3. User Interface Overview

3.1. Menu Bar

- **File** – Open image, save individual images, save all images, save settings, reset to defaults, exit.
- **Settings** – Opens the appearance and fonts settings dialog.
- **Help** – Displays program information (About).

3.2. Status Bar (bottom)

- **Coordinates** – shows cursor position in pixels and in μm (after calibration).
- **Calibration** – shows the current $\text{px}/\mu\text{m}$ (Or px/mm , px/in).
- **Zoom slider** – range 10–500% (100% = 1:1). Affects the currently selected view tab.
- **Fit to Window** button – fits the image to the window size in the active tab.

3.3. Zoom and Pan

- **Ctrl + mouse wheel** – zoom in/out in the active view.
- **Ctrl + left mouse button + drag** – pan the image when zoomed in.
- **Fit to Window** – resets view to full fit.

3.4. Interaction Modes

- **Length measurement drawing** – activated by the **Measure Length (Draw)** button in the **Thickness** tab.
- **ROI drawing** – activated by the **Enable ROI Selection** button in the **Porosity** tab (after creating a new ROI).
- **Calibration line drawing** – activated by the **Enable Calibration Line** button in the **Calibration** tab.
- **Hardness diagonal drawing** – activated by the **Set Diagonal 1/2** buttons in the **Hardness** tab (calibration required).

Only one mode can be active at a time. Mode buttons act as toggles (highlighted when active).

4. Loading Images

4.1. From File

- **Menu File** → **Open Image** (or shortcut Ctrl+O).
- Supported formats: PNG, JPG, JPEG, TIFF, BMP, GIF, WEBP.
- The image is added to the clipboard (Clipboard tab) and automatically loaded into analysis (Processed tab).

4.2. Using the Clipboard

The **Clipboard** tab stores a list of images (120×90 thumbnails). Functions:

- **Open Image** – add an image from disk to the clipboard (without loading into analysis).
- **Use in Analysis** – load the selected image into analysis (resets measurements, ROI, but keeps calibration and processing settings).
- **Delete Selected** – remove the selected image from the clipboard.
- **Clear All** – remove all images.

After loading into analysis, the image appears in the **Original**, **Processed**, **Thickness**, etc. tabs. The image name is displayed on the tab labels (e.g., my_image_Original).

4.3. Live Camera Capture

The **Camera** tab provides live USB camera preview and frame capture.

Camera setup:

1. Click **Refresh** – scans cameras (indices 0–4) using DirectShow.
2. Select a camera from the list.
3. Click **Start** – live preview appears in the **Live** tab.
4. Set **Display Mode** (Color/Grayscale) and adjustments (brightness, contrast, gamma) – changes are immediate.
5. To capture a frame, click **Capture Image** – the image is added to the clipboard (Clipboard tab) with name CapturedImage.
6. Click **Use in Analysis** – loads the captured image into analysis.

Note: Live preview only works in the **Live** tab. Stopping the camera (**Stop** button) stops the preview.

5. Image View Tabs

Each tab uses its own `ImageViewer` instance supporting zoom/pan and optional annotations (measurement lines, ROI, calibration, scale bar). Depending on the tab, some annotations are enabled/disabled.

Tab	Displays	Annotations (lines, ROI, calibration, scale bar)	Drawing allowed
Live	Live camera image	None (only image)	No
Clipboard	List of thumbnails	–	–
Original	Original image (no adjustments)	Measurements, ROI, scale bar (no calibration)	No
Processed	Image after adjustments & filters	All (measurements, ROI, calibration, scale bar)	Yes
Thickness	Processed image	Only length measurements and scale bar	Yes
ROI	Processed image	Only ROI outlines and scale bar	No
Binary	Binary image (inverted)	Only ROI outlines and scale bar	No
Fusion	Fusion image (colored pores by diameter)	ROI outlines, scale bar, porosity legend	No

Annotations include:

- Length measurement lines (with arrowheads and labels).
- Calibration line (green, with endpoint circles).

- Hardness diagonals (different colors for diag1 and diag2).
- ROI outlines (red while drawing, green when finished, with label).
- Scale bar (yellow) with length label.
- Porosity legend (only in **Fusion** tab).

All annotations are dynamically drawn when zooming – their line thickness and font size scale appropriately for readability.

6. Control Panel

The right panel contains tabs grouping the functions.

6.1. Camera

Described in section 4.3. Additional:

- **Brightness** (-100..100), **Contrast** (0..200%), **Gamma** (0.1..5.0) – real-time adjustments.
- **Display Mode** – toggles color/grayscale (only affects preview, not captured image).
- After **Capture Image**, the **Use in Analysis** button becomes active.

6.2. Adjustments

Pre-processing before binarization. Changes affect the **Processed**, **Thickness**, **ROI** tabs, and indirectly **Binary** and **Fusion** (after re-binarization).

Parameters:

- **Display Mode** – color/grayscale (affects the processed image).
- **Brightness** (-100..100) – added to pixel values.
- **Contrast** (0..200%) – contrast multiplier.
- **Gamma** (0.1..5.0) – gamma correction.
- **Basic Filters** (Gaussian, Median, Bilateral, Erosion, Dilation) – each in range 0..15 (0..5 for erosion/dilation). Value indicates kernel size (odd) or iteration count.

Note: Changing any parameter automatically refreshes the processed image (with 100 ms debounce). Binarization is not automatic – you must click **Whole Image** or **Selected ROI**.

6.3. Thickness (Length Measurements)

Described in detail in chapter 9. Contains:

- **Measure Length (Draw)** – drawing mode toggle.
- List of saved measurements.
- Buttons: **Add Current Line**, **Delete Selected**, **Clear All Lengths**.
- **Selected:** field and statistics (average, standard deviation, count).
- **Show Measurements Table** – opens a separate table window.

6.4. Porosity (Porosity and ROI)

The most extensive tab. Contains:

ROI Management:

- ROI table (columns: Name, Type, Status, Binarization, Porosity (%)).
- Buttons: **New ROI**, **Delete ROI**, **Finish ROI**, **Clear ROI**.
- Shape selection: Rectangle, Ellipse, Polygon, Freeform.
- **Enable ROI Selection** – toggles ROI drawing mode on the **Processed** image.
- **Average Porosity** – average porosity over all finished ROIs.

Binarisation:

- Drop-down list of methods (see chapter 7).
- **Method Settings...** button – opens additional settings for the selected method.
- **Low threshold / High threshold** sliders (visible only for Manual Range).
- **Adaptive Block Size** slider (for Adaptive single scale).
- **Multi-Otsu classes** and **Pore class index** spinboxes (for Multi-Otsu).
- **Show threshold range on Processed image** checkbox – for Manual Range, overlays a semi-transparent red layer on pixels within the threshold range (helps threshold selection).
- Execution buttons: **Whole Image**, **Selected ROI**, **All ROIs (union)**.
- **Current Porosity** label – shows porosity from the last binarization.
- **Show Detailed Porosity Results** – opens the `PorosityResultsDialog` window.
- **Save Results** section – buttons for reports and images.
- **Legend position** – selects legend placement on Fusion image (Left bottom, Center bottom, Right bottom).

Notes:

- Binarization always runs on the **Processed** image (after adjustments and filters).

- After changing binarization settings (method, thresholds, ROI), there is a 200 ms delay then automatic re-binarization for the current ROI/whole image, and the result appears in **Binary** and **Fusion** tabs.
- Porosity for ROI is calculated using the ROI mask and shown in the table after clicking **Recalculate all porosities** (automatic after each binarization).

6.5. Hardness

Described in chapter 11. Contains:

- Method selection: **Vickers** or **Knoop**.
- Load selection (standard list or **Custom Load** field in grams).
- **Set Diagonal 1** / **Set Diagonal 2** – activate drawing of the respective diagonal on the **Processed** image.
- **Clear Diagonals** – removes both diagonals.
- **Calculate Hardness** – computes hardness from current diagonals.
- Table with measurement history.
- **Hardness Detailed Table** – opens a separate window with statistics.

6.6. Calibration

Described in chapter 12. Contains:

- **Enable Calibration Line** – toggles calibration line drawing mode.
- Unit (μm , mm, in) and **Known length** field.
- **Calibrate** button – after drawing the line, calculates $\text{px}/\mu\text{m}$.
- **Multiple Calibrations** – list of saved calibrations (max 5). Buttons: Add Current, Apply Calibration, Delete Selected.
- **Save Calibration to File** / **Load Calibration from File** – save/load a single calibration to/from JSON.
- **Load calibration at application start** – if checked, the last used calibration is loaded on startup.
- **Scale Bar** section – allows adding a scale bar to the image (requires calibration).

7. Binarization Methods & Settings

Binarization converts a grayscale image into a binary image where white (255) represents pores (areas of interest) and black (0) background. The application offers 12 methods.

7.1. Available Methods (list order)

Index	Method Name	Description
0	Manual Range (low-high)	Manual lower and upper thresholds (0-255). Pixels in the range become white.
1	Otsu	Global Otsu thresholding (inverted).
2	Triangle	Triangle method (inverted).
3	Adaptive (single scale)	Adaptive Gaussian thresholding with a single block size (odd 3-5).
4	Watershed	Marker-based segmentation (watershed) – detects pore boundaries.
5	Multi-Otsu	Multi-class Otsu – select the class corresponding to pores (index 0 is darkest).
6	Fusion Voting (5 methods)	Majority voting of five methods (Otsu, Triangle, Yen, Li, IsoData).
7	Yen	Yen thresholding.
8	Li	Li thresholding (minimum cross-entropy).
9	IsoData	Iterative IsoData thresholding.
10	Minimum	Minimum thresholding.
11	Multi-scale Adaptive	Logical OR of three adaptive binarizations with different block sizes (11,25,51).

Important: All methods (except Manual Range and Multi-Otsu) assume pores are darker than background. If pores are brighter, invert the image before binarization (no direct option – use external editor or manual range with reversed thresholds).

7.2. Method Settings Dialog

Clicking **Method Settings...** opens a separate dialog that adapts to the selected method:

- **For Manual Range (0):**
 - Low/High sliders (same as main panel).
 - **Pick low from image** and **Pick high from image** buttons – after clicking, the program enters threshold picking mode. Click on the **Processed** image (or on the histogram palette) – the pixel value becomes the threshold.
 - Histogram display with marked thresholds.
 - Clickable grayscale palette (0–255 bar) – click to set threshold.
 - **Show threshold range on Processed image** checkbox (duplicate of main panel).
- **For Adaptive single scale (3):**
 - **Block size** slider (only in this dialog).
- **For Multi-Otsu (5):**
 - **Number of classes** (2–5) and **Pore class index** (0..classes–1) spinboxes.
- **For other methods** – the dialog shows only a description and a Close button.

This dialog is modeless – you can keep it open while working.

7.3. Threshold Range Preview on Processed Image

When Manual Range is selected and the **Show threshold range** checkbox is checked, the **Processed** image gets a semi-transparent red overlay on pixels that are **NOT** within the low–high range (i.e., will be treated as background). This helps visual threshold tuning. The preview updates with 50 ms delay after threshold changes.

8. ROI Management (Regions of Interest)

ROIs allow porosity calculation only in selected areas. Maximum **5 ROIs**.

8.1. Creating a New ROI

1. Click **New ROI** – a new row appears in the table, named **ROI x**. It becomes the current ROI.
2. Choose a shape (Rectangle, Ellipse, Polygon, Freeform) via radio buttons.
3. Click **Enable ROI Selection** – toggles drawing mode (button text changes to **Drawing ...**).
4. Draw on the **Processed** image (details in 8.2).
5. When finished, click **Finish ROI** – the shape is closed (for polygon/freeform, closing can also happen automatically).

6. The ROI status becomes "Finished" in the table.

8.2. Drawing Shapes

- **Rectangle / Ellipse** – click and drag to define the rectangle/ellipse. Release to create the shape (but not finished – click **Finish ROI**).
- **Polygon** – click successive points around the area. To close the polygon, click on the first point (or hold Shift near the first point) or use **Finish ROI**. A temporary line from the last point to the cursor is shown.
- **Freeform** – drag the mouse to draw a freehand contour. Release the mouse – the contour is automatically closed (connected to the start point) and the ROI is finished (no need for **Finish ROI**).

Tips:

- While drawing, the outline is red (for unfinished ROIs).
- After finishing, the outline turns green and a label with the ROI name appears.
- To delete the current ROI, click **Delete ROI**. To clear the shape (without deleting the row) – **Clear ROI**.

8.3. ROI Table

- **Name** – editable (double-click the cell).
- **Type** – displays the shape.
- **Status** – "Drawing" or "Finished".
- **Binarization** – shows the currently selected binarization method (informational only).
- **Porosity (%)** – after binarization (using **Selected ROI** or **All ROIs**), the value is filled for that ROI.

Clicking a row selects that ROI as current. Then **Selected ROI** binarization applies to it, and on the Fusion image only pores inside that ROI are shown (if mask is used). The porosity label at the top also refers to the current ROI.

8.4. Calculating Porosity for ROI

- **Selected ROI** – performs binarization only within the current ROI (cropping to the mask). Binary and Fusion tabs show only the ROI area.
- **All ROIs (union)** – combines masks of all finished ROIs and calculates porosity in the union.
- **Whole Image** – ignores ROI, calculates for the whole image.

After each binarization, the ROI table updates porosity values (queried in background by `PorosityWorker`). The average porosity over ROIs is displayed above the table.

9. Length Measurements

Up to 10 length measurements can be stored.

9.1. Drawing Mode

1. Go to the **Thickness** tab (or **Processed** – where drawing is also allowed).
2. Click **Measure Length (Draw)** – button text changes to `Drawing measurement...`.
3. On the image, click and drag to draw a line. Release – the line is automatically added to the list (if under limit 10). Alternatively, you can later click **Add Current Line** if drawing mode is off but a temporary line exists.
4. Lines are drawn with arrowheads at both ends and a label like `M1: 12.34  $\mu\text{m}$` in the middle.

Limit: Max 10 measurements. Exceeding shows a warning.

9.2. Managing Measurements

- **Measurement list** – shows entries `Meas 1: 12.34  $\mu\text{m}$` . Clicking an entry selects it.
- **Delete Selected** – removes the selected measurement.
- **Clear All Lengths** – removes all.
- **Add Current Line** – if you drew a line but didn't add it (or drawing mode was off), you can add it manually.
- **Statistics** – update after every change: average, standard deviation, count. Values are converted to the current length unit (from calibration). If calibration unit is mm or inches, results are displayed accordingly.

9.3. Results Table

Click **Show Measurements Table** – opens a separate `MeasurementsDialog` with all measurements and a statistics row. The dialog updates automatically when measurements change.

Note: Length measurements are independent of binarization – they work on the processed image but do not require binarization. Calibration is necessary for real-world units.

10. Area Measurements

The application automatically calculates areas for drawn ROIs (rectangle, ellipse, polygon, freeform). There is no separate drawing mode – area is computed as soon as an ROI is finished (after clicking **Finish ROI**).

Accessing results:

- In Word/Excel reports, the **Area measurements** section lists all ROIs and their areas (in squared units according to calibration).
- For each ROI, additional details are given (for rectangle: width × height; for ellipse: diameters; for polygon/freeform: perimeter).

Note: Area measurements are not shown in a separate dialog; they are only available in reports. To see them, generate a Word/Excel report.

11. Hardness Measurement (Vickers / Knoop)

Hardness is calculated from diagonal lengths of an indentation. Calibration is required (knowledge of $\text{px}/\mu\text{m}$).

11.1. Selecting Method and Load

- In the **Hardness** tab, choose **Vickers** or **Knoop**.
- Select a standard load from the list or enter a custom value in **Custom Load** (grams). If both are provided, custom takes precedence.

11.2. Drawing Diagonals

1. Click **Set Diagonal 1** – button highlights blue. Drawing mode for diagonal 1 is activated.

2. On the **Processed** image, click and drag to draw a line corresponding to the first diagonal of the indentation. Release – the line is saved (cyan color).
3. Repeat for **Set Diagonal 2** (magenta color).
4. Use **Clear Diagonals** to remove both.

Verification: Labels show diagonal lengths in μm (after calibration). For Vickers, the ratio of the two diagonals must not exceed 1.5. Minimum diagonal length is 20 μm (safety).

11.3. Calculating and Saving Results

1. Click **Calculate Hardness**.
2. The program computes hardness using formulas:
 - Vickers: $\text{HV} = 1.8544 * (F / d^2)$ where F in kg, d = average of both diagonals in mm.
 - Knoop: $\text{HK} = 14.229 * (F / d^2)$ where d = longer diagonal in mm.
3. The result appears in the **Result** field, e.g., 123.4 HV (or HK).
4. The measurement is added to the table in the Hardness tab.
5. Multiple measurements can be stored (different indentations).
6. Click **Hardness Detailed Table** – opens a separate window with all measurements and statistics (average, standard deviation, count).

Note: Only one pair of diagonals is active at a time. Drawing new diagonals overwrites the previous pair. The measurement history stores copies.

12. Scale Calibration

Calibration is essential to convert pixels to real units (μm , mm, in). The program can store up to 5 calibrations.

12.1. Drawing the Calibration Line

1. In the **Calibration** tab, click **Enable Calibration Line** – the button text changes, drawing mode activated.
2. On the **Processed** image, locate an object of known length (e.g., a micrometer scale). Click and drag to draw a line exactly on that segment.
3. Release – the line becomes a temporary green line. The **Status** field changes to "Calibration line drawn. Press Calibrate."

12.2. Entering Known Length and Calibrating

1. Choose the unit (μm , mm, in) from the **Unit** list.
2. Enter the **Known length** value (e.g., 100 μm).
3. Click **Calibrate**.
4. The program computes $\text{pixels_per_micron} = \text{pixel length} / \text{real length in } \mu\text{m}$ (converting mm/in accordingly). The result is displayed in the **Calibration:** field, e.g., 2.34 $\text{px}/\mu\text{m}$.
5. Calibration becomes active immediately. The status bar updates.

12.3. Managing Multiple Calibrations

- **Add Current** – saves the current calibration (pixels_per_micron and unit) to the list (max 5). Prompts for a label.
- **Apply Calibration** – loads the selected calibration from the list and sets it active.
- **Delete Selected** – removes the selected calibration from the list.
- Calibrations are stored in `multi_calibration.json` in the program directory.

12.4. Saving/Loading to/from JSON

- **Save Calibration to File** – saves the *current* calibration (not the list) to a JSON file.
- **Load Calibration from File** – loads a calibration from a JSON file and sets it active (does not add to the multi-calibration list).

12.5. Auto-load Calibration at Startup

Check **Load calibration at application start** – on next launch, the last used calibration (from list or from `settings.ini`) will be loaded. The state is saved.

13. Scale Bar

A scale bar is drawn on images in the **Processed**, **Thickness**, **ROI**, **Binary**, and **Fusion** tabs (not on Original or Live).

Adding a scale bar:

1. Ensure calibration is active.
2. In the **Calibration** tab, under **Scale Bar**:

- Enter desired length (e.g., 50).
 - Choose unit (μm , mm, in) – can be different from calibration unit (program converts).
 - Click **Add Scale Bar**.
3. The bar appears at the bottom right corner of the image (90% width from left, 90% height from top). It is yellow, with a label like 50 μm .
 4. To remove, click **Remove Scale Bar**.

Note: The bar's pixel length is computed from the current calibration. If calibration changes, the bar may disappear – add it again.

14. Generating Reports (Word/Excel)

The application can generate detailed reports in Word (.docx) and Excel (.xlsx). The required libraries (python-docx and openpyxl) are bundled in the .exe.

14.1. Selecting Report Sections

Before saving, a **Select Report Sections** dialog appears, allowing you to choose which sections to include:

- Processing parameters
- Porosity per ROI
- Pore size distribution (detailed)
- Length measurements
- Area measurements
- Hardness measurements
- Images

All are checked by default.

14.2. Data Preparation

The program collects:

- **Processing parameters** – brightness, contrast, gamma, filters, binarization method, thresholds, calibration, scale bar.
- **Porosity per ROI** – table with names, types, status, porosity.
- **Pore size distribution** – for whole image and each ROI: categories by thresholds (default 10,30,60,100 μm), count, min, max, average diameter, average area. Histograms are generated as temporary PNG files.

- **Length measurements** – list with units and statistics.
- **Area measurements** – for each ROI (type, area, details).
- **Hardness measurements** – method, load, value.
- **Images** – original (no annotations), processed (with annotations), thickness (with length measurements), ROI (with outlines), binary (with ROI), fusion (colored pores + legend). For each ROI, additional binary and fusion cropped images.

14.3. Background Generation

After choosing the save path, the report is generated in a separate thread (ReportWorker). A progress dialog appears ("Generating Word/Excel report...") with a cancel option. On completion, a success or error message is shown.

14.4. Report Contents

Word Report (.docx):

- Title page (Material Analyzer - Full Report), date, image name.
- For each selected section: headings, tables, histogram charts, images (inserted as inline shapes, width 4 inches or 3 inches for smaller ones).
- Fonts follow application settings (default Arial 11 pt for normal text, larger for headings).

Excel Report (.xlsx):

- Separate worksheets for each section: Processing Parameters, Porosity per ROI, Pore Distribution - Whole, Pore Distribution - <ROI name>, Length Measurements, Area Measurements, Hardness Measurements.
- In pore distribution worksheets, each category in a separate row.
- No images or charts (only tabular data).

Note: Reports are generated asynchronously to avoid UI freezing. Generation can be cancelled (though the thread may leave an incomplete file).

15. Saving Images

15.1. Saving a Single Image

- In the **Porosity** tab, under **Save Results**, there is a **Save Image** dropdown with choices: Original, Processed, Thickness, ROI, Binary, Fusion.
- Click **Save** – a file save dialog opens. Default filename: `image_name_type.png`.
- For Processed, Thickness, ROI images, the saved version includes all annotations (lines, ROI, scale bar, etc.) – i.e., what you see on screen.
- Formats: PNG, JPG, TIFF (selectable in the save dialog).

15.2. Saving All Images at Once

- **Menu File → Save All Images...**
 - A dialog appears with checkboxes for each image type.
 - After selection and OK, choose a destination folder.
 - Each selected image is saved as `image_name_type.png` in that folder. Annotated versions are saved.
-

16. Customizing Appearance & Default Settings

The **Settings** dialog (menu **Settings** → **Appearance Settings**) allows personalization of colors, fonts, line widths, and other parameters.

16.1. Colors and Line Widths (Visual Objects tab)

For each of the following, you can change the color (**Change Color**) and line width in pixels (**Change Line Width**):

- Measurements (length)
- Hardness diagonal 1
- Hardness diagonal 2
- Calibration line
- ROI (drawing)
- ROI (finished)
- Scale bar
- Temporary line

Changes apply immediately to all views.

16.2. Porosity Legend Colors (Porosity Legend tab)

- Five ranges (Range1..Range5) – each with its own color.
- Default colors: red, green, orange, blue, magenta.
- Additionally, **Background Color** (default black) and **Text Color** (default white).

16.3. Legend Scale and Position

- **Legend font scale** (0.2–5.0) – multiplier for legend text size (relative to auto-scaled).
- **Legend box scale** (0.5–3.0) – multiplier for the entire legend box size.
- **Legend position** (Left bottom, Center bottom, Right bottom) – placement on the Fusion image.

16.4. Interface Fonts (Fonts tab)

You can change the font for:

- Labels
- Buttons
- Tables
- Results (porosity, statistics)
- Annotations (on images – default Arial 16)
- ROI labels (default Arial 16 bold)
- Measurement values (default Arial 12)
- Hardness labels (Arial 12)
- Scale bar labels (Arial 11)

After changing, the interface is refreshed.

16.5. Other Settings (Other tab)

- **Endpoint marker size** (pixels) – size of circles at line ends (calibration line, diagonals) – default 8.
- **Word report font size** (8–40) – base font size for Word reports.
- **Report image annotation font size** (8–48) – font size for annotations added to images in reports (e.g., ROI names on binary/fusion images).

16.6. Saving Current Settings as Default

- **Save Current Settings as Default** – saves all current parameters (colors, fonts, binarization settings, calibration?) as defaults, overwriting `settings.ini`.
- **Reset to Default Settings** – restores factory values.

Note: Settings are automatically saved on normal application exit. Manual “Save as Default” allows you to keep preferences for future sessions.

17. Keyboard Shortcuts

Shortcut	Action
<code>Ctrl+O</code>	Open image (file) – adds to clipboard and loads into analysis.
<code>Ctrl+S</code>	Save image from the current tab.
<code>Ctrl+D</code>	Open detailed porosity results dialog (PorosityResultsDialog).
<code>Ctrl+M</code>	Toggle length measurement drawing mode.
<code>Ctrl+R</code>	Create a new ROI.
<code>Ctrl+Z</code>	Undo last length measurement or last area measurement (removes list).
<code>Ctrl + mouse wheel</code>	Zoom in/out in active view.
<code>Ctrl + left mouse button + drag</code>	Pan the image when zoomed in.

18. Error Handling and Logging

The application logs messages to `material_analyzer.log` in the program directory. Log levels: INFO and ERROR.

Common errors and causes:

- *Cannot open camera* – camera not available or in use by another application.
- *Binarization failed* – error in selected method (e.g., invalid parameters). Check the method and thresholds.

- *Diagonal ratio > 1.5* – for Vickers, the difference between diagonals is too large.
- *Diagonal too short* – diagonal shorter than 20 μm .
- *No calibration* – attempting measurement or calibration without active calibration.
- *Maximum 10 measurements reached* – length measurement limit.

If the application crashes unexpectedly, please send the log file to technical support.

19. Frequently Asked Questions (FAQ)

Q: Why doesn't the Fusion image update after changing the binarization method?

A: After changing the method (or thresholds), binarization is automatically triggered with a 200 ms delay. If it doesn't happen, manually click **Whole Image** or **Selected ROI**.

Q: How do I rename an ROI?

A: In the ROI table, double-click the name cell and type a new name.

Q: Can I save the Fusion image without the legend?

A: The legend is an integral part of the Fusion image. To save without legend, use the **Processed** or **Binary** tab.

Q: What do the pore colors on the Fusion image mean?

A: Colors correspond to pore diameter ranges (default: red $\leq 10 \mu\text{m}$, green 10–30 μm , orange 30–60 μm , blue 60–100 μm , magenta $> 100 \mu\text{m}$). Ranges can be changed in the detailed porosity dialog (**Show Detailed Porosity Results**).

Q: How can I add my own binarization method?

A: There is no built-in extension mechanism. However, you can use **Fusion Voting** or **Multi-scale Adaptive** which combine several approaches.

Q: Why does the scale bar disappear after calibration?

A: The scale bar is removed when calibration changes (for safety). Add it again using **Add Scale Bar**.

Q: Can I measure the area of an arbitrary region without creating an ROI?

A: Not directly. You can create a temporary ROI, measure the area (via report), then delete the ROI.

Q: Are there limits on the number of ROIs or measurements?

A: Max 5 ROIs, max 10 length measurements (intentional to keep UI responsive).

20. Technical Support

For issues, bugs, or suggestions, please contact:

- **Email:** mgoral@hub.pl (author: Marek Goral)

Please include in your report:

- The `material_analyzer.log` file
- The image that caused the issue (if applicable)
- Steps to reproduce the error
- Your operating system version

Thank you for using **Coating Analyzer Binek 2.4 Beta** – we wish you successful material analyses!

Manual prepared based on source code version 2.4 Beta. Last updated: 2026-05-22